

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Pipeline Processing

Lecture No.- 01

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Recap of Previous Lecture



Topic

Disk Access Time

Topic

Disk Addressing

Topics to be Covered



Topic

Pipeline Processing



[NAT]

Ans = 19

$$1 \text{ sector capacity} = \frac{20 \text{ MB}}{20} = 1 \text{ MB}$$

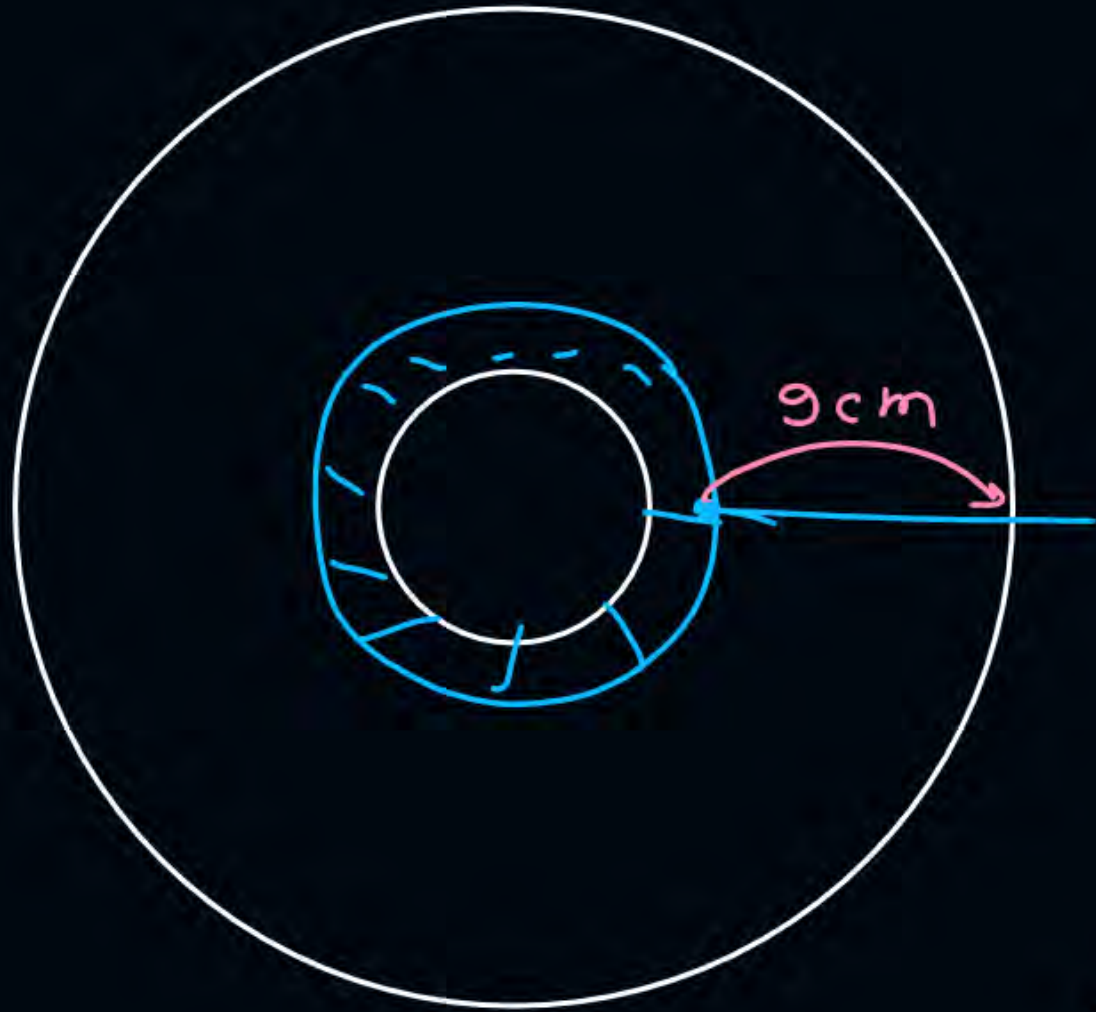


#Q. A disk has 10 equidistant tracks. The diameters of the innermost and outermost tracks are 2 cm and 20 cm respectively. The track has a storage capacity of 20MB. If the disk has 20 sectors per track and is currently at the start of the 5th sector of the inner-most track and the head can move at a speed of 10 meters/sec and it is rotating at constant angular velocity of 3000 RPM, how much time will it take to read 1 MB contiguous data starting from the sector 3 of the outer-most track?

$$\begin{aligned} \text{for } 1000 \text{ cm time} &= 1 \text{ sec} \\ \text{for } 9 \text{ cm time} &= \frac{1 \text{ sec}}{1000 \text{ cm}} * 9 \text{ cm} \\ &= 9 \text{ msec} \end{aligned}$$

$$1 \text{ rotation time} = \frac{60000}{3000} = 20 \text{ ms}$$

$$\begin{aligned} 1 \text{ sector rotation time} &= \frac{20 \text{ ms}}{20} \\ &= 1 \text{ ms} \end{aligned}$$



while 9 ms of seek, disk rotates
9 sectors.



when seek time over, R/W head
will be at start of $5+9=14^{\text{th}}$ sector.



rotational latency = 9 sectors (from 14^{th}
rotation time to 3^{rd})
 $= 9 \text{ ms}$

$$\begin{aligned} \text{Total time} &= 9 \text{ ms} + 9 \text{ ms} + \frac{20 \text{ ms}}{20} \\ &= \underline{\underline{19 \text{ ms}}} \quad \text{Ans.} \end{aligned}$$

#Q. Consider a hard disk with 16 recording surfaces (0-15) having 16384 cylinders (0-16383) and each track contains 64 sectors (0-63). Data storage capacity of in each sector is 512 Bytes. Data are organized cylinder-wise and addressing format is <cylinder no., surface no., sector no.>. A file of size 42797 KB is stored in the disk and the starting disk location of the file is <1200, 9, 40>. What is the cylinder number of the last sector of the file, if it is stored in a contiguous manner?

$$\text{Ans} = 1284$$



Topic : Parallel Processing

Parallel Processing: Simultaneous data processing

Types:

- Vector Processing
- Array Processing
- Pipeline Processing (Pipelining) ✓



Topic : Flynn's Classification of Computers

1. SISD :- single instⁿ stream, single Data stream $\Rightarrow$ sequential
2. SIMD :- ---||--- multiple ---||--- $\Rightarrow$ Pipeline
3. ~~MISD~~ :- multiple ---||--- single ---||---
4. MIMD multiple ---||--- multiple ---||--- $\Rightarrow$ multiple pipeline
(super scalar processor)



Topic : Pipeline Processing

- Pipelining is useful, When same processing is applied over multiple inputs





Topic : Pipeline Processing

- Technique to decompose a sequential process into sub-operations
- Sub-operations are performed in **segments** (*stages, phases*)
- Each segment performs its suboperation in parallel on different inputs
- **Task:** One operation performed in all segments



Topic : Pipelining Example



shirt making $\Rightarrow$ cut stitch finish pack



Topic : Example

9 min

10 min

8 min

7 min

10 min

Clock pulse number	Segment 1 <i>cut</i>	Segment 2 <i>stitch</i>	Segment 3 <i>finish</i>	Segment 4 <i>pack</i>
1 10 min	S1			
2 20	S2	S1		
3 30	S3	S2	S1	
4 40	S4	S3	S2	S1
5 50	S5	S4	S3	S2
6 60	—	S5	S4	S3
7 70	—	—	S5	S4
8 80	—	—	—	S5

inputs	non-pipeline	pipeline
$n=5$	$5 * 40 = 200 \text{ min}$	80 min
$n=6$	$6 * 40 = 240 \text{ min}$	90 min
$n=7$	$7 * 40 = 280 \text{ min}$	100 min



Topic : Pipeline Cycle Time

(t_p)



min. amount of time in which all segments can perform their respective suboperations.



Topic : General Consideration About Pipeline

- Consider a k segment pipeline with clock cycle time = t_p to perform n tasks

	<u>cycles</u>	<u>time</u>
Time required to perform 1 st task	k	$k * t_p$
remaining $(n-1)$ tasks	$n-1$	$(n-1)t_p$
Total time for all n tasks = $(k+n-1)$		$(k+n-1)t_p$



Topic : General Consideration About Pipeline



- Consider a non-pipeline system that takes t_n time to perform a task

time needed to perform n tasks = $n * t_n$



Topic : General Consideration About Pipeline

- Performance of a pipeline is given by **Speed up ratio**.

$$S = \frac{n * t_n}{(k+n-1) t_p}$$

as no. of inputs increases
 $n \gg k$

ignore $k-1$

$$S_{\text{ideal or max}} = \frac{t_n}{t_p}$$

ideal condition:-

initial time needed to fill pipe
($k-1$ cycles) is ignored.



each cycle one output from
pipeline



Topic : General Consideration About Pipeline

special case:-

if to process one input, pipeline and non-pipeline take equal time.

$$t_n = k * t_p$$

then

$$S_{ideal} = k$$

Practically:-

$$t_n \leq k t_p$$

- #Q. A non-pipeline system takes 50 ns to process a task. The same task can be processed in a six-segment pipeline with a clock cycle of 10ns.
1. Determine the speedup ratio of the pipeline for 100 tasks.
 2. What is the maximum speedup that can be achieved?

$$t_n = 50 \text{ ns}$$

$$k = 6$$

$$t_p = 10 \text{ ns}$$

$$n = 100$$

$$1. \quad S = \frac{100 * 50}{(6+100-1)10} = 4.76$$

$$2. \quad S_{ideal} = \frac{50}{10} = 5$$



Topic : Synchronous Pipeline

only in pipeline $\leftarrow$ Intermediate buffer or reg.



$$t_p = \max \text{ of seg. delays} = \max(9, 12, 10, 11) = 12 \text{ ns}$$

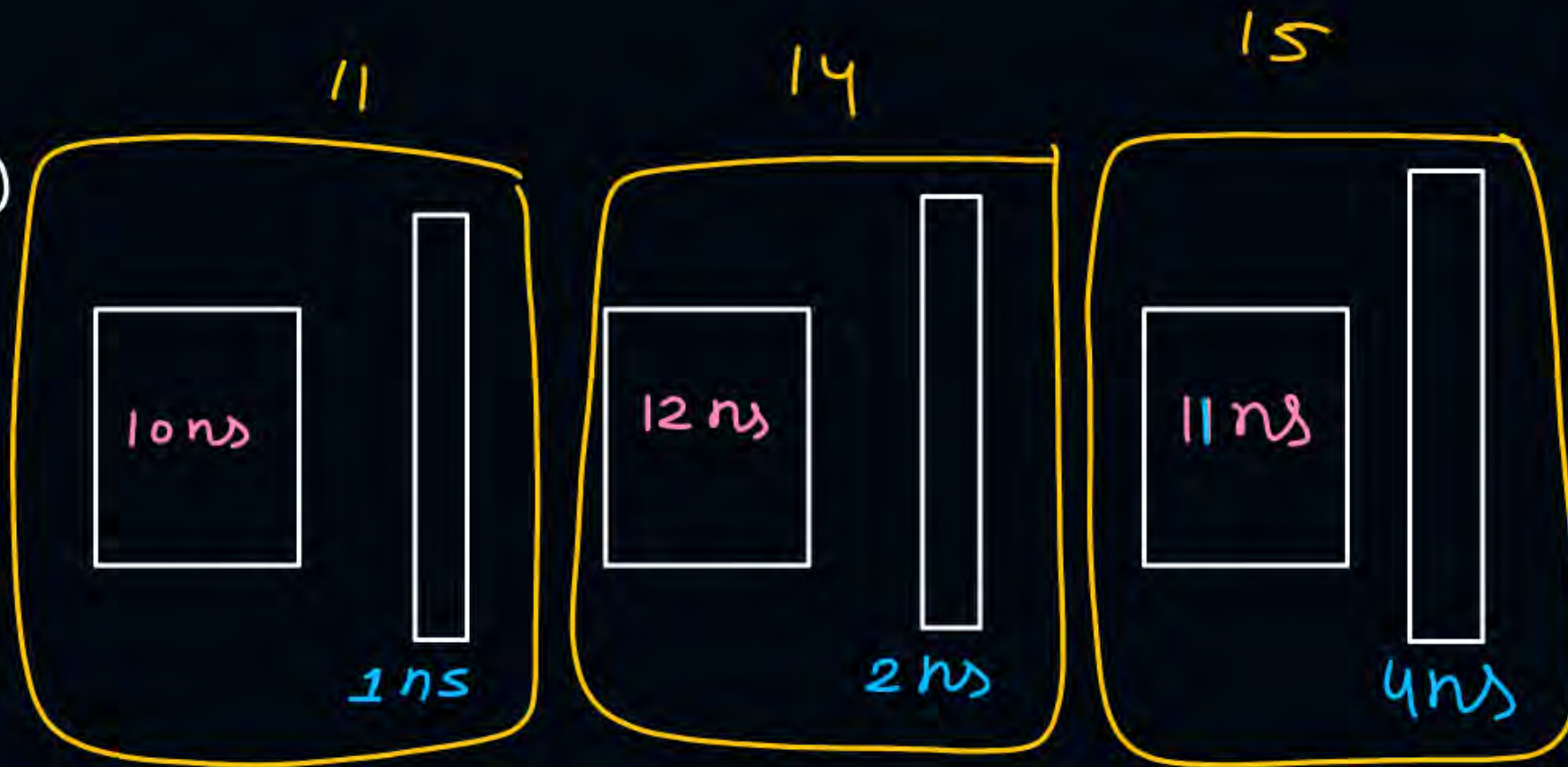
$$t_p = \max \text{ of seg delay} + \text{reg. delay} = 13 \text{ ns}$$



Topic : Cycle Time in Synchronous Pipeline

$$\begin{aligned}t_n &= \text{sum of all seg. delays} \\&= 9 + 12 + 10 + 11 \\&= 42 \text{ ns}\end{aligned}$$

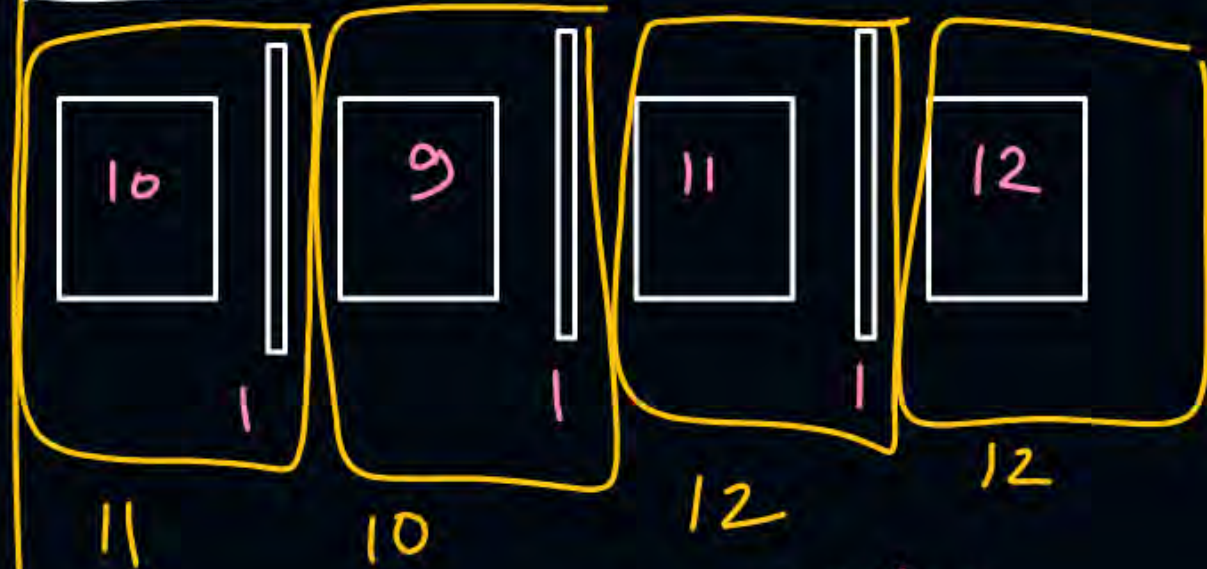
Ques)



$$t_p = 15\text{ ns}$$

Ques) 4 seg. delays 10, 9, 11, 12 ns
buffer used b/w the segments.
and delay of each buffer 1 ns.

$$t_p = \text{---} ?$$



$t_p = 12\text{ ns}$ for test series
 $t_p = 13\text{ ns}$ for Gate exam.

#Q. Consider 6 segment pipeline with segment delay of segments as 20ns, 26ns, 21ns, 21ns, 24ns and 28ns respectively. Calculate processing time of pipeline for 1000 tasks?

$$t_p = 28 \text{ ns}$$

$$\begin{aligned} \text{pipeline time} &= (6 + 1000 - 1) 28 \text{ ns} \\ &= 28140 \text{ ns} \end{aligned}$$



Happy Learning

THANK - YOU

